

Preliminary note on Conjecture 3.2

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Current understanding

Let M be the initial phase space, let e denote the Euclidean metric, and let $g_t = \phi_t^* e$ be the time-dependent pullback metric generated by a volume-preserving flow. On $\mathbb{M}_0 = [0, \tau] \times M$,

$$G_{0,a}(t, x) = \begin{pmatrix} a^{-2} & 0 \\ 0 & g_t(x) \end{pmatrix}, \quad \Delta_{G_{0,a}} F(t, \cdot) = a^2 \partial_{tt} F(t, \cdot) + \Delta_{g_t} F(t, \cdot).$$

Let S_a, H_a be the Sobolev and Cheeger constants on $(\mathbb{M}_0, G_{0,a})$, and let s_D, h_D be the dynamic constants on M . The known result is

$$H_a = S_a \leq s_D = h_D,$$

with H_a, S_a nondecreasing in a . Conjecture 3.2 asks whether

$$\lim_{a \rightarrow \infty} S_a = s_D, \quad \lim_{a \rightarrow \infty} H_a = h_D,$$

and whether minimizers converge to the time-copied dynamic minimizers. The inequality $S_a \leq s_D$ is supplied by testing S_a on time-independent functions $F(t, x) = f(x)$. A possible way to view the remaining functional step is to prove a liminf lower bound, using compactness to show that near-minimizers become almost independent of time as $a \rightarrow \infty$.

Possible first target

I would first try to prove $S_a \rightarrow s_D$ in a simple setting, for example on a compact boundaryless manifold or flat torus. Since $S_a \leq s_D$ is known, the key point is

$$\liminf_{a \rightarrow \infty} S_a \geq s_D.$$

After cancelling the common volume factor in $dV_{G_{0,a}} = a^{-1} dt d\ell$, take a near-minimizing sequence U_a . Since the quotient is invariant under adding constants and multiplying by nonzero scalars, one may replace U_a by the normalized representative

$$F_a := \frac{U_a - \alpha_a}{D(U_a)}, \quad D(U_a) := \inf_{\alpha \in \mathbb{R}} \int_0^\tau \int_M |U_a - \alpha| d\ell dt,$$

where α_a is a median or minimizing constant. Then F_a is still near-minimizing and satisfies

$$\inf_{\alpha \in \mathbb{R}} \int_0^\tau \int_M |F_a - \alpha| d\ell dt = 1.$$

In the cancelled quotient, uniform boundedness of the numerator

$$\int_0^\tau \int_M (a^2 |\partial_t F_a|^2 + \|\nabla_{g_t} F_a(t, \cdot)\|_{g_t}^2)^{1/2} d\ell dt \leq C$$

gives

$$\int_0^\tau \int_M |\partial_t F_a| d\ell dt \leq C/a \rightarrow 0.$$

Thus a one-dimensional Poincare inequality in time should make F_a close in $L^1([0, \tau] \times M)$ to $\bar{F}_a(x) = \tau^{-1} \int_0^\tau F_a(t, x) dt$.

The step I am not yet sure about is the appropriate lower semicontinuity statement for the spatial total variation, roughly of the form

$$\liminf_{a \rightarrow \infty} \int_0^\tau \int_M \|\nabla_{g_t} F_a(t, \cdot)\|_{g_t} d\ell dt \geq \int_0^\tau \int_M \|\nabla_{g_t} f\|_{g_t} d\ell dt,$$

where $F_a \rightarrow f(x)$ in L^1 after passing to a subsequence. If the denominator also passes to the limit correctly, this would imply the desired liminf inequality. If this works, $H_a \rightarrow h_D$ follows from $H_a = S_a$ and $s_D = h_D$. The convergence of minimizers could then be treated as a subsequent step after the functional convergence is understood.

Questions where guidance would be valuable

- Is Conjecture 3.2 still open, and is $S_a \rightarrow s_D$ a reasonable first target?
- Are BV compactness and lower semicontinuity for weighted total variation the right tools, or is there a more natural route?
- Would the Dirichlet analogue, Conjecture 3.5, be a cleaner starting point, or does it introduce additional boundary difficulties?

If this first functional step is reasonable, I would try to write a more detailed note identifying exactly where compactness and lower semicontinuity are needed, before attempting to extend the argument toward the full conjecture.